Tema 1 21.03.2025

Modified Booth

X = -107 = 1001 0101

Y = 85 = 0101 0101

P = -9095

-M = 1010 1011

COUNT	OVR	A	Q[8]	Q	R	M
000	0	0000 0000	1	1001 0101	0	0101 0101
	+	0101 0101				
		0101 0101				-M
		0010 1010	1	1100 1010	0	1010 1011
001	0	0001 0101	0	1110 0101	0	
010	⊕	0101 0101				
	0	0110 1010				
		0011 0101	0	0111 0010	0	
011	0	0001 1010	1	0011 1001	0	
100	⊕	0101 0101				
	0	0110 1111				
		0011 0111	1	1001 1100	0	
101	0	0001 1011	1	1100 1110	0	
110	0	0000 1101	1	1110 0111	0	-M
111	0 ⊕	1010 1011				
		1011 1000				
		1101 1100	0	1111 0011	1	

$P = 1101\ 1100\ 0111\ 1001_{C_2} = 1010\ 0011\ 1000\ 0111_{S+}$

$= -(8192 + 512 + 256 + 128 + 4 + 2 + 1) = -9095$

Radix 4 Booth ⟶ se shiftează cu 2 poziții

$X = -78 = 1011\ 0010$

$Y = -93 = 1\,1010\ 0011$

$P = (-78)\cdot(-93) = 7254$

CNT	A	Q	Q[-1]	M
00	0 0000 0000	1011 0010	0	Y 110100011
⊕	0 1011 1010		-2M	-Y
	0 1011 1010			001011101
	→ RSHIFT			
	0 0010 1110	1010 1100	1	2Y
			+M	101000110
01 ⊕	1 1010 0011			-2Y
	1 1101 0001			010110110
	1 1111 0100	0110 1011	0	
			-M	
10 ⊕	0 0101 1101			
	0 0101 0001			
	0 0001 0100	0101 1010	1	
			-M	
11 ⊕	0 0101 1101			
	0 0111 0001			
	0 0001 1100	0101 0110	1	

$P = 2 + 4 + 16 + 64 + 1024 + 2048 + 4096 = 7254$

Radix 8 Booth

CNT	A	Q	Q[-1]	M
00	00 0000 0000	1 1011 0010	0	M 11 1010 0011
⊕	11 0100 0110		+2M	
	11 0100 0110			−M
	11 1110 1000	1 1011 0110	0	00 0101 1101
01 ⊕	00 1011 1010		-2M	2M 11 0100 0110
	00 1010 0010			-2M 00 1011 1010
	00 0001 0100	0 1011 0110	1	
			-M	
⊕	00 0101 1101			
10	00 0111 0001			
	00 0000 1110	0 0101 0110	1	

$P = 4096 + 2048 + 1024 + 64 + 16 + 4 + 2 = 7254$